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Rf-Technologies

Fire damper with motorised actuator CU-LT 200x100



ISSUED 11/10/2024
VALID UNTIL 11/10/2029

THIRD PARTY VERIFIED
in accordance with ISO 14025, EN 15804+A2
and B-EPD-PCR (Version 18.10.2022)

FUNCTIONNAL UNIT AND MODULES DECLARED

(cradle-to-gate with options)

1 CU-LT 200x100 fire damper with a motorised actuator, to be used in conjunction with partitions to maintain fire compartments in heating, ventilation and air conditioning installations, with a fire resistance up to 120 minutes and a reference service life of 30 years

A123	A4	A5	B	C	D
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The intended use of this EPD is to communicate scientifically based environmental information for construction products, for the purpose of assessing the environmental performance of buildings. This EPD is only valid when registered on www.b-epd.be. The FPS Public Health cannot be held responsible for the information provided by the owner of the EPD.

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1 PRODUCT DESCRIPTION

1.1 Product name

Fire damper CU-LT 200x100 with motorised actuator

1.2 Product description and intended use

CU-LT 200x100 is an optimised rectangular fire damper with a fire resistance up to 120 minutes. It is the smallest type of a fire damper series and has a width of 200 mm and a height of 100 mm. A minimal pressure loss is guaranteed thanks to the thin blade and the transmission located outside the tunnel. A galvanised steel tunnel contributes to the light weight of the damper. The fire damper can be equipped with various types of mechanisms (motorised or not), depending on the specific needs linked to the project or to the local regulations. In this EPD, one specific motorised actuator is considered as a reference.

Fire dampers are installed where air ducts penetrate fire-resistant compartment walls. Their role is to restore the fire resistance grade of the penetrated wall and to prevent smoke propagation. Fire dampers are distinguished by their degree of fire resistance, by their aeraulic properties and by their installation ease. Rf-Technologies solutions are designed and approved for all types of walls: vertical and horizontal, in concrete or lightweight materials. They are compliant with the most stringent local and European standards. Furthermore, the products for the compartmentation and smoke control provide fire safety for commercial and institutional buildings across Europe: office buildings, hospitals, museums, multi-family housing, public buildings and shopping centres.

In general, a CU-LT fire damper consists of three main parts, i.e. a fire damper casing, mainly composed of a galvanised steel duct, a fire damper blade, composed of a 25 mm thick Promatect H plate, and an actuator (motorised or not). In this EPD, a motorised actuator is considered as a reference.

This EPD is a specific EPD for one specific product, i.e. CU-LT 200x100 fire damper with motorised actuator, from one single company, i.e. Rf-Technologies, that is produced in one single site, i.e. Oosterzele, Belgium.

1.3 Reference flow / declared unit

The functional unit for this EPD is 1 CU-LT 200x100 fire damper with a motorised actuator, to be used in conjunction with partitions to maintain fire compartments in heating, ventilation and air conditioning installations, with a fire resistance up to 120 minutes and a reference service life of 30 years.

Packaging of the raw materials and the finished product is included.

The net weight per reference flow is 4,856 kg.



1.4 Installation

The product can be installed in different varieties, in respect with the Declaration of Performance (DoP):

- Installation in a rigid wall and floor;
- Installation in flexible wall (metal stud gypsum plasterboard wall);
- Installation in flexible wall (metal stud gypsum plasterboard wall), sealing with gypsum;
- Installation in flexible wall (metal stud gypsum plasterboard wall), sealing with mortar;
- Installation in shaft wall, sealing with gypsum;
- Installation in shaft wall, sealing with rigid rock wool boards with coating;
- Installation in shaft wall with IFW installation kit and plaster joint filler;
- Installation in flexible and rigid wall, sealing with rigid rock wool boards with coating;
- Installation in rigid floor, sealing with rigid rock wool boards with coating;
- Installation remote from the wall, sealing and insulation with rigid rock wool boards with coating;
- Installation remote from the wall + GEOFLAM plates (threaded rods and U-shaped steel profiles).

Because of the different ways of installation, the product is declared as '*as produced*' in this EPD.

Consequently, regarding installation, this EPD only includes the environmental impact related to the product itself (i.e. treatment of the waste arising from the packaging of the final product). The impact of possible additional products and materials, necessary for the installation of the product, is not included in this EPD. However, this impact shall be taken into account at building level.

More detailed information on the different installation scenarios can be found in the chapter "*Additional technical information for scenario development*" in this EPD.



1.5 Composition and content

In general, a CU-LT fire damper consists of three main parts, i.e. a fire damper casing, a fire damper blade and an actuator. The damper casing is composed of a galvanised steel duct, steel flanges, steel brackets and aluminium rivets. The fire damper blade is made of a 25 mm thick Promatect H plate, which pivots on a steel hinge. The fire damper is activated with an actuator, of which both motorised and non-motorised types are available for this product. In this EPD, one specific motorised actuator is considered as a reference. The composing elements of the CU-LT 200x100 fire damper with motorised actuator are detailed in the table below.

Components	Composition / content / ingredients	Quantity
CU-LT 200x100 fire damper with motorised actuator	Casing	42%
	Blade	12%
	Mechanics	14%
	Motorised actuator	32%

The product does not contain materials listed in the "Candidate list of Substances of Very High Concern for authorization".

1.6 Reference service life

The reference service life (RSL) for a CU-LT 200x100 fire damper with motorised actuator is estimated at 30 years problem-free and maintenance-free. The product does not require specific maintenance nor replacements over its lifetime.

The proof of the product specification is done through the Declaration of Performance (DoP). This document is evaluated by a notified body. The CE marking of the product is based on specific endurance tests, where 10.000 cycles of the motorised actuator are proven. In function of the test frequency of the product by the user, these cycles can be spread over 30 years. After 30 years, the actuator could be replaced and the fire damper itself could be retained.

Furthermore, building practice in Belgium shows that public buildings are often renovated and completely stripped down to the shell after 30 years. In that case, the fire damper with motorised actuator will also be removed from the building.

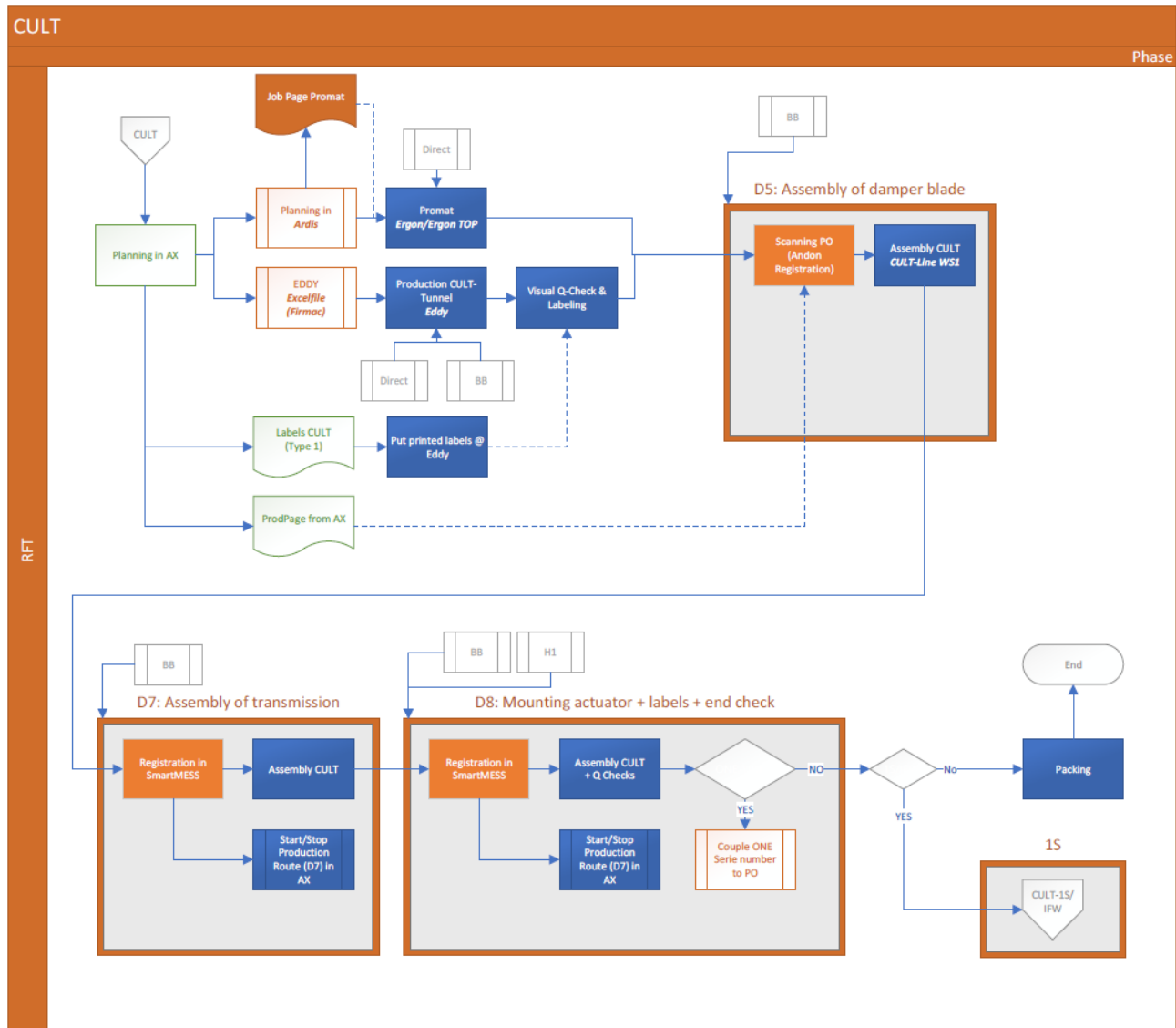
1.7 Description of geographical representativity

The CU-LT 200x100 fire damper with motorised actuator is produced in Belgium by the Belgian company Rf-Technologies and is installed in Belgium. The EPD is representative for the Belgian market. The composed datasets for this EPD are representative and relevant for a CU-LT 200x100 fire damper with motorised actuator produced by Rf-Technologies in Belgium.



1.8 Description of the production process and technology

For the production of the CU-LT 200x100 fire damper, at first, the parts for the calcium silicate damper blade and the metal tunnel body are produced as sub-assembly parts. These two parts come together in the final assembly line. Here, the following steps are performed: assembly of the damper blade, assembly of the transmission, mounting of the actuator on the damper, adding labels, end control and test. Finally, the finished and tested fire damper is packed.



2 TECHNICAL DATA / PHYSICAL CHARACTERISTICS

The technical characteristics of the CU-LT 200x100 fire damper with motorised actuator are given in the Declaration of Performance (DOP) of the product. This document can be found on the website of the producer through the following link: [Geoptimaliseerde rechthoekige brandklep tot 120' \(rft.eu\)](https://www.rf-t.nl/geoptimaliseerde-rechthoekige-brandklep-tot-120-rft.eu)

DECLARATION OF PERFORMANCE

CE_DoP_Rf-t_C3_EN = 1-05/2023

Unique identification code of the product-type:			CU-LT		
2. Intended use/s:			Rectangular fire damper to be used in conjunction with partitions to maintain fire compartments in heating, ventilating and air conditioning installations.		
3. Manufacturer:			Rf-Technologies NV, Lange Ambachtstraat 40, B-9860 Oosterzele		
4. System/s of AWCP:			System 1		
5. Harmonised standard / European Assessment Document; notified body / European Technical Assessment, Technical Assessment Body, notified body; certificate of constancy of performance:			EN 15650:2010, BCCA with identification number 0749; BCCA-0749-CPR-BC1-606-0464-15650.05-0464-2822-UKCA-CPR-0060		
6. Declared performance according to EN 15650:2010			(Fire resistance according to EN 1366-2 and classifications according to EN 13501-3)		
Essential characteristics					
Range	Wall type	Wall	Sealing	Installation	Performance
200x100 mm ≤ CU-LT ≤ 800x600 mm	Rigid wall	Aerated concrete ≥ 100 mm	Mortar	1	EI 90 (v, i ↔ o) S - (500 Pa)
			Gypsum	1	EI 120 (v, i ↔ o) S - (500 Pa)
			Stone wool + coating ≥ 140 kg/m³ + coated casing	1	EI 120 (v, i ↔ o) S - (300 Pa)
			Stone wool + coating ≥ 140 kg/m³	1	EI 90 (v, i ↔ o) S - (300 Pa)
			Galvanised duct + stone wool + coating ≥ 140 kg/m³ 1x60 mm + IFW installation kit	2	EI 60 (v, i ↔ o) S - (300 Pa)
			Galvanised duct + stone wool + coating ≥ 140 kg/m³ 1x80 mm + IFW installation kit	2	EI 90 (v, i ↔ o) S - (300 Pa)
			Galvanised duct + stone wool + coating ≥ 140 kg/m³ 2x50 mm + IFW installation kit	2	EI 90 (v, i ↔ o) S - (300 Pa)
			Galvanised duct + GEOFLAM® F 45 mm + mortar	2	EI 120 (v, i ↔ o) S - (500 Pa)
			Galvanised duct + GEOFLAM® Light 35 mm + mortar	2	EI 120 (v, i ↔ o) S - (500 Pa)
	Flexible wall	Metal studs gypsum plasterboard Type A (EN 520) ≥ 100 mm	Installation kit	3	EI 60 (v, i ↔ o) S - (500 Pa)
			Gypsum	1	EI 60 (v, i ↔ o) S - (500 Pa)
			Stone wool + coating ≥ 140 kg/m³	1	EI 60 (v, i ↔ o) S - (300 Pa)
			Galvanised duct + stone wool + coating ≥ 140 kg/m³ 1x60 mm + IFW installation kit	2	EI 60 (v, i ↔ o) S - (300 Pa)
			Galvanised duct + stone wool + coating ≥ 140 kg/m³ 2x50 mm + IFW installation kit	2	EI 60 (v, i ↔ o) S - (300 Pa)
		Metal studs gypsum plasterboard Type F (EN 520) ≥ 100 mm	Installation kit	3	EI 90 (v, i ↔ o) S - (500 Pa)
			Gypsum	1	EI 90 (v, i ↔ o) S - (500 Pa)
			Mortar	4	EI 90 (v, i ↔ o) S - (300 Pa)
			Stone wool + coating ≥ 140 kg/m³ + coated casing	1	EI 120 (v, i ↔ o) S - (500 Pa)
			Stone wool + coating ≥ 140 kg/m³	1	EI 90 (v, i ↔ o) S - (300 Pa)
			Galvanised duct + stone wool + coating ≥ 140 kg/m³ 2x50 mm + IFW installation kit	2	EI 90 (v, i ↔ o) S - (300 Pa)
	Asymmetrical flexible wall (shaft wall)	Metal studs gypsum plasterboard Type A (EN 520) ≥ 75 mm	Gypsum	5	EI 30 (v, i ↔ o) S - (500 Pa)
		Metal studs gypsum plasterboard Type F (EN 520) ≥ 75 mm	Stone wool + coating ≥ 140 kg/m³	5	EI 30 (v, i ↔ o) S - (300 Pa)
		Metal studs gypsum plasterboard Type F (EN 520) ≥ 90 mm	Installation kit	6	EI 90 (v, i ↔ o) S - (300 Pa)
	Rigid floor	Reinforced concrete ≥ 110 mm	Mortar	1	EI 90 (v, i ↔ o) S - (500 Pa)
		Reinforced concrete ≥ 150 mm	Gypsum	1	EI 120 (v, i ↔ o) S - (500 Pa)
		Aerated concrete ≥ 150 mm	Stone wool + coating ≥ 140 kg/m³ + coated casing	1	EI 120 (v, i ↔ o) S - (300 Pa)
			Stone wool + coating ≥ 140 kg/m³	1	EI 90 (v, i ↔ o) S - (300 Pa)
1	Type of installation: built-in 0°/90°/180°/270°. Minimal distances authorised.			2	Type of installation: remote from the wall, 0°/180°. Minimal distances authorised.
3	Type of installation: built-in 0°/90°/180°/270°.			4	Type of installation: built-in 0°/180°. Minimal distances authorised.
5	Type of installation: built-in 0°/180°. Minimal distances authorised.			6	Type of installation: built-in 0°/180°. Minimal distances authorised.
Nominal activation conditions/sensitivity:					
Response delay (response time): closure time			Pass		
Operational reliability: cycling			Pass		
Durability of response delay:			MFUSP - 50 cycles; MMAG - 300 cycles; BFL(T) - 10000 cycles; ONE - 10000 cycles; ONE-X - 10000 cycles; UNIQ - 10000 cycles		
Durability of operational reliability:			Pass		
Protection against corrosion according to EN 60068-2-52:			Pass		
Damper casing leakage according to EN 1751:			Pass		
			≥ class C		
The performance of the product identified above is in conformity with the set of declared performance/s. This declaration of performance is issued, in accordance with Regulation (EU) No 305/2011, under the sole responsibility of the manufacturer identified above.					
				Signed for and on behalf of the manufacturer by: Mathieu Steensels, Technical Manager	

Harmonised standard
EN 15650:2010

Signed for and on behalf of the manufacturer by:
Mathieu Steenland, Technical Manager

Oosterzele, 03/2023



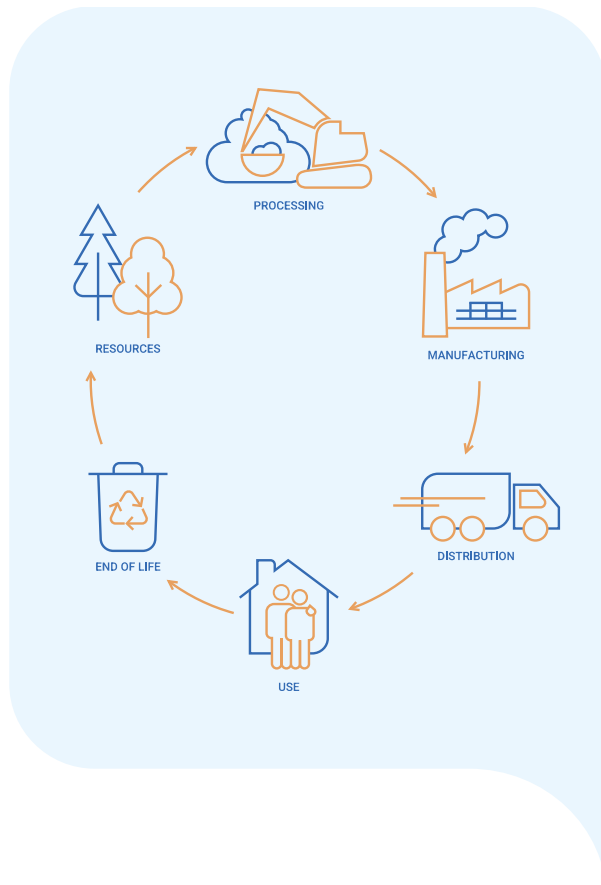
Furthermore, additional information and options for this product can be consulted on the website of the producer through the following link: [Geoptimaliseerde rechthoekige brandklep tot 120' \(rft.eu\)](https://www.rf-t.nl/geoptimaliseerde-rechthoekige-brandklep-tot-120-rft.eu)



3 LCA STUDY

3.1 Date of LCA study

August 2024



3.2 Software

For the calculation of the LCA results, the software program SimaPro 9.6.0.1 (PréConsultants, 2024) has been used.

3.3 Information on allocation

At Rf-Technologies, different types of fire damper are produced. Only facility level data were available for infrastructure and land use. These data have been allocated to the analysed product by dividing the total amounts by the total annual production volume (physical relationship). For the other production data, specific data (electricity consumption, ancillary materials, packaging of final product

and production waste) or theoretical data (transport means) were available.

3.4 Information on cut-off

The following processes are considered to be below cut-off:

- Packaging of raw materials with mass contribution <3% in module A1;
- Transportation to production site of raw materials with mass contribution <3% in module A2;
- Transportation of packaging of raw materials to production site in module A2;
- Transportation of the ancillary materials to the production site in module A3;
- Packaging of the ancillary materials in module A3.

In all cases, it is assumed that the cut-off criteria of EN 15804+A2 are met.

3.5 Information on excluded processes

Following processes were excluded from the inventory: Environmental impacts caused by the personnel of the production plants, e.g. waste from the cafeteria and sanitary installations, accidental pollution caused by human mistakes, or environmental effects caused by commuter traffic.



3.6 Information on biogenic carbon modelling

The CU-LT 200x100 fire damper with motorised actuator does not contain any biogenic carbon. However, the accompanying packaging does contain biogenic carbon.

Biogenic carbon content	(kg C / FU)
Biogenic carbon content in product (at the gate)	0,000
Biogenic carbon content in accompanying packaging (at the gate)	0,809

3.7 Information on carbon offsetting

Carbon offsetting is not allowed in the EN 15804 and hence not taken into account in the calculations.

3.8 Additional or deviating characterisation factors

For the characterisation factors, the package 'EN 15804 based on EF3.1 from the EC-JRC website' has been used. No additional or deviating characterisation factors were used.



3.9 Specificity

The data used for the LCA are specific for the CU-LT 200x100 fire damper with motorised actuator, which is manufactured by Rf-Technologies in its factory at Oosterzele, Belgium.

3.10 Period of data collection

Manufacturer specific data have been collected for the year 2022.

3.11 Information on data collection

Company-specific data for the product stage have been collected by Rf-Technologies and were provided to Enperas NV through a data collection questionnaire. Enperas NV uses publicly available generic data for all background processes, such as the production of electricity, transportation by means of a specific truck, etc. Primary data is used for modules A1, A2 and A3. The rest of the study is based on scenarios (module A4, module A5, modules C1-C4 and module D).

3.12 Database used for background data

The main LCI sources used in this LCA study are the Ecoinvent v3.9.1 Cut-off (Wernet et al., 2023) and the Industry Data 2.0 databases.

3.13 Energy mix

The Belgian electricity mix (consumption mix + import) has been used to model electricity use in life cycle stages A3 and D. The used data record in module A3 is the Ecoinvent v3.9.1

Cut-off data record '*Electricity, medium voltage {BE} / market for electricity, medium voltage / Cut-off, U*' (Wernet et al., 2016). For the own produced solar energy, used in module A3, the Ecoinvent v3.9.1 Cut-off data record '*Electricity, low voltage {BE} / electricity production, photovoltaic, 3kWp slanted-roof installation, single-Si, panel, mounted / Cut-off, U*' is used. The data record used in module D for electricity recovery grace to incineration of waste is the Ecoinvent v3.9.1 Cut-off data record '*Electricity, high voltage {BE} / market for electricity, high voltage / Cut-off, U_no transmission network*'.

For other energy recovery grace to incineration of waste in life cycle stage D, a European data record from the Ecoinvent v3.9.1 Cut-off database is used, i.e. '*Heat, district or industrial, natural gas {Europe without Switzerland} / heat production, natural gas, at industrial furnace >100kW / Cut-off, U*'.



4 PRODUCTION SITES

The production site is located at Lange Ambachtstraat 40 in 9860 Oosterzele, Belgium.

5 SYSTEM BOUNDARIES

Product stage			Construction installation stage		Use stage							End of life stage				Beyond the system boundaries
Raw materials	Transport	Manufacturing	Transport	Construction installation stage	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
☒	☒	☒	☒	☒	☐	☐	☐	☐	☐	☐	☐	☒	☒	☒	☒	☒

X = included in the EPD

□ = module not declared

In module A5, only the impact of the treatment of the packaging waste arising from the final product is considered in order to keep mass balances correct. Installation losses and installation impacts are not considered due to the different installation ways for this product.



6 POTENTIAL ENVIRONMENTAL IMPACTS PER REFERENCE FLOW

CU-LT 200x100 fire damper with motorised actuator

The results in the following table are valid for 1 CU-LT fire damper 200x100 with motorised actuator. The reference flow is 4,856 kg per piece.

	Production			Construction process stage		Use stage							End-of-life stage				D Reuse, recovery, recycling
	A1 Raw material	A2 Transport	A3 Manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	
 GWP total (kg CO2 equiv/FU)	1,76E+01	2,60E-01	-1,13E+00	1,22E-01	2,56E+00	MND	MND	MND	MND	MND	MND	MND	0,00E+00	4,05E-02	7,83E-02	9,88E-02	-4,38E+00
 GWP fossil (kg CO2 equiv/FU)	1,78E+01	2,59E-01	9,55E-01	1,22E-01	1,36E-01	MND	MND	MND	MND	MND	MND	MND	0,00E+00	4,05E-02	7,79E-02	9,87E-02	-3,08E+00
 GWP biogenic (kg CO2 equiv/FU)	-2,47E-01	8,75E-05	-2,09E+00	4,07E-05	2,43E+00	MND	MND	MND	MND	MND	MND	MND	0,00E+00	1,29E-05	2,65E-04	3,83E-05	-1,29E+00
 GWP luluc (kg CO2 equiv/FU)	2,11E-02	1,27E-04	4,74E-03	5,86E-05	1,51E-05	MND	MND	MND	MND	MND	MND	MND	0,00E+00	1,97E-05	9,43E-05	1,25E-05	-1,43E-03
 ODP (kg CFC 11 equiv/FU)	4,06E-07	5,88E-09	8,54E-08	2,70E-09	8,91E-10	MND	MND	MND	MND	MND	MND	MND	0,00E+00	8,81E-10	1,23E-09	4,96E-10	-8,64E-08
 AP (mol H+ eq/FU)	2,49E-01	6,39E-04	5,16E-03	2,73E-04	2,86E-04	MND	MND	MND	MND	MND	MND	MND	0,00E+00	1,32E-04	9,41E-04	1,55E-04	-6,07E-02
 EP - freshwater (kg P equiv/FU)	1,46E-03	2,18E-06	6,15E-05	9,94E-07	3,21E-07	MND	MND	MND	MND	MND	MND	MND	0,00E+00	3,24E-07	3,96E-06	3,67E-07	-2,55E-04
 EP - marine (kg N equiv/FU)	2,12E-02	1,70E-04	1,39E-03	6,89E-05	1,25E-04	MND	MND	MND	MND	MND	MND	MND	0,00E+00	4,49E-05	2,04E-04	1,58E-04	-3,92E-03
 EP - terrestrial (mol N equiv/FU)	4,21E-01	1,78E-03	1,65E-02	7,19E-04	1,40E-03	MND	MND	MND	MND	MND	MND	MND	0,00E+00	4,80E-04	2,35E-03	5,74E-04	-5,09E-02
 POCP (kg NMVOC equiv/FU)	9,01E-02	1,04E-03	5,67E-03	4,33E-04	4,12E-04	MND	MND	MND	MND	MND	MND	MND	0,00E+00	1,97E-04	6,97E-04	2,19E-04	-2,27E-02
 ADP Elements (kg Sb equiv/FU)	1,87E-03	7,38E-07	8,35E-06	3,78E-07	1,08E-07	MND	MND	MND	MND	MND	MND	MND	0,00E+00	1,30E-07	5,50E-06	3,40E-07	-7,49E-04
 ADP fossil fuels (MJ/FU)	2,48E+02	3,93E+00	2,09E+01	1,77E+00	4,33E-01	MND	MND	MND	MND	MND	MND	MND	0,00E+00	5,74E-01	1,05E+00	4,22E-01	-3,28E+01
 WDP (m³ water eq deprived /FU)	6,83E+00	1,87E-02	5,48E-01	7,48E-03	3,50E-03	MND	MND	MND	MND	MND	MND	MND	0,00E+00	2,34E-03	1,38E-02	6,70E-03	-1,02E+00

GWP TOTAL = TOTAL GLOBAL WARMING POTENTIAL (CLIMATE CHANGE); GWP-LULUC = GLOBAL WARMING POTENTIAL (CLIMATE CHANGE) LAND USE AND LAND USE CHANGE; ODP = OZONE DEPLETION POTENTIAL; AP = ACIDIFICATION POTENTIAL FOR SOIL AND WATER; EP = EUTROPHICATION POTENTIAL; POCP = PHOTOCHEMICAL OZONE CREATION; ADPE = ABIOTIC DEPLETION POTENTIAL – ELEMENTS; ADPF = ABIOTIC DEPLETION POTENTIAL – FOSSIL FUELS; WDP = WATER USE (WATER (USER) DEPRIVATION POTENTIAL, DEPRIVATION-WEIGHTED WATER CONSUMPTION)

7 RESOURCE USE

CU-LT 200x100 fire damper with motorised actuator

The results in the following table are valid for 1 CU-LT fire damper 200x100 with motorised actuator. The reference flow is 4,856 kg per piece.

	Production			Construction process stage		Use stage							End-of-life stage				
	A1 Raw material	A2 Transport	A3 Manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	D Reuse, recovery, recycling
PERE (MJ/FU, net calorific value)	3,31E+01	5,74E-02	1,84E+01	2,79E-02	2,27E+01	MND	MND	MND	MND	MND	MND	MND	0,00E+00	8,76E-03	1,79E-01	2,66E-02	-1,86E+01
PERM (MJ/FU, net calorific value)	2,94E+00	0,00E+00	1,97E+01	0,00E+00	-2,27E+01	MND	MND	MND	MND	MND	MND	MND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
PERT (MJ/FU, net calorific value)	3,61E+01	5,74E-02	3,82E+01	2,79E-02	1,34E-02	MND	MND	MND	MND	MND	MND	MND	0,00E+00	8,76E-03	1,79E-01	2,66E-02	-6,42E+00
PENRE (MJ/FU, net calorific value)	2,21E+02	3,93E+00	1,85E+01	1,77E+00	2,75E+00	MND	MND	MND	MND	MND	MND	MND	0,00E+00	5,74E-01	1,05E+00	4,22E-01	-3,36E+01
PENRM (MJ/FU, net calorific value)	2,74E+01	0,00E+00	2,43E+00	0,00E+00	-2,32E+00	MND	MND	MND	MND	MND	MND	MND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	8,53E-01
PENRT (MJ/FU, net calorific value)	2,48E+02	3,93E+00	2,09E+01	1,77E+00	4,33E-01	MND	MND	MND	MND	MND	MND	MND	0,00E+00	5,74E-01	1,05E+00	4,22E-01	-3,28E+01
SM (kg/FU)	1,24E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	MND	MND	MND	MND	MND	MND	MND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
RSF (MJ/FU, net calorific value)	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	MND	MND	MND	MND	MND	MND	MND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
NRSF (MJ/FU, net calorific value)	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	MND	MND	MND	MND	MND	MND	MND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
FW (m³ water eq/FU)	4,64E-01	5,77E-04	1,50E-02	2,47E-04	3,52E-04	MND	MND	MND	MND	MND	MND	MND	0,00E+00	7,66E-05	5,36E-04	4,08E-04	-2,85E-02

PERE = USE OF RENEWABLE PRIMARY ENERGY EXCLUDING RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PERM = USE OF RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PERT = TOTAL USE OF RENEWABLE PRIMARY ENERGY RESOURCES; PENRE = USE OF NON-RENEWABLE PRIMARY ENERGY EXCLUDING NON-RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PENRM = USE OF NON-RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PENRT = TOTAL USE OF NON-RENEWABLE PRIMARY ENERGY RESOURCES; SM = USE OF SECONDARY MATERIAL; RSF = USE OF RENEWABLE SECONDARY FUELS; NRSF = USE OF NON-RENEWABLE SECONDARY FUELS; FW = NET USE OF FRESH WATER

8 WASTE CATEGORIES & OUTPUT FLOWS

CU-LT 200x100 fire damper with motorised actuator

The results in the following table are valid for 1 CU-LT fire damper 200x100 with motorised actuator. The reference flow is 4,856 kg per piece.

	Production			Construction process stage		Use stage							End-of-life stage				
	A1 Raw material	A2 Transport	A3 Manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	
Hazardous waste disposed (kg/FU)	5,16E-03	2,44E-05	8,63E-05	0,00E+00	2,51E-06	MND	MND	MND	MND	MND	MND	MND	0,00E+00	3,66E-06	5,71E-06	1,95E-06	-3,44E-04
Non-hazardous waste disposed (kg/FU)	5,22E+00	3,38E-01	3,54E-01	0,00E+00	2,57E-02	MND	MND	MND	MND	MND	MND	MND	0,00E+00	3,66E-06	5,71E-06	1,95E-06	-2,28E-02
Radioactive waste disposed (kg/FU)	3,89E-04	1,22E-06	7,05E-05	6,13E-07	6,00E-07	MND	MND	MND	MND	MND	MND	MND	0,00E+00	1,87E-07	2,42E-06	6,97E-07	-2,10E-06
Components for re-use (kg/FU)	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	MND	MND	MND	MND	MND	MND	MND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
Materials for recycling (kg/FU)	0,00E+00	0,00E+00	3,07E-01	0,00E+00	7,45E-01	MND	MND	MND	MND	MND	MND	MND	0,00E+00	0,00E+00	3,47E+00	0,00E+00	0,00E+00
Materials for energy recovery (kg/FU)	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	MND	MND	MND	MND	MND	MND	MND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
Exported energy (MJ/FU)	0,00E+00	0,00E+00	4,01E-01	0,00E+00	3,14E+00	MND	MND	MND	MND	MND	MND	MND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00

9 IMPACT CATEGORIES ADDITIONAL TO EN 15804

CU-LT 200x100 fire damper with motorised actuator

The results in the following table are valid for 1 CU-LT fire damper 200x100 with motorised actuator. The reference flow is 4,856 kg per piece.

	Production			Construction process stage		Use stage							End-of-life stage				
	A1 Raw material	A2 Transport	A3 Manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	
 PM (disease incidence)	1,67E-06	2,52E-08	7,99E-08	9,49E-09	3,75E-09	MND	MND	MND	MND	MND	MND	MND	0,00E+00	3,21E-09	1,25E-08	3,06E-09	-2,43E-07
 IRHH (kg U235 eq/FU)	8,32E-01	1,92E-03	9,14E-02	9,36E-04	7,58E-04	MND	MND	MND	MND	MND	MND	MND	0,00E+00	2,88E-04	3,16E-03	9,39E-04	-9,38E-03
 ETF (CTUe/FU)	3,33E+02	1,89E+00	6,90E+00	8,76E-01	2,73E-01	MND	MND	MND	MND	MND	MND	MND	0,00E+00	2,83E-01	8,38E-01	5,18E-01	-4,31E+01
 HTCE (CTUh/FU)	8,12E-08	1,16E-10	3,28E-09	5,35E-11	2,02E-10	MND	MND	MND	MND	MND	MND	MND	0,00E+00	1,84E-11	1,30E-10	1,60E-11	2,29E-08
 HTnCE (CTUh/FU)	1,56E-06	2,81E-09	1,21E-08	1,23E-09	6,81E-10	MND	MND	MND	MND	MND	MND	MND	0,00E+00	4,04E-10	6,05E-09	5,96E-10	-7,83E-07
 Land Use Related impacts (dimensionless)	1,18E+02	3,92E+00	2,17E+02	1,26E+00	2,30E-01	MND	MND	MND	MND	MND	MND	MND	0,00E+00	3,42E-01	2,04E+00	8,78E-01	-5,26E+01

HTCE = HUMAN TOXICITY – CANCER EFFECTS; HTNCE = HUMAN TOXICITY – NON CANCER EFFECTS; ETF = ECOTOXICITY – FRESHWATER; (POTENTIAL COMPARATIVE TOXIC UNIT)

PM = PARTICULATE MATTER (POTENTIAL INCIDENCE OF DISEASE DUE TO PM EMISSIONS);

IRHH = IONIZING RADIATION – HUMAN HEALTH EFFECTS;

TS (POTENTIAL HUMAN EXPOSURE EFFICIENCY RELATIVE TO U235);

9.1 Environmental impact categories explained

	Global Warming Potential	The global warming potential of a gas refers to the total contribution to global warming resulting from the emission of one unit of that gas relative to one unit of the reference gas, carbon dioxide, which is assigned a value of 1. It is split up in 4: – Global Warming Potential total (GWP-total) which is the sum of GWP-fossil, GWP-biogenic and GWP-luluc – Global Warming Potential fossil fuels (GWP-fossil) : The global warming potential related to greenhouse gas (GHG) emissions to any media originating from the oxidation and/or reduction of fossil fuels by means of their transformation or degradation (e.g. combustion, digestion, landfilling, etc). – Global Warming Potential biogenic (GWP-biogenic) : The global warming potential related to carbon emissions to air (CO₂, CO and CH₄) originating from the oxidation and/or reduction of aboveground biomass by means of its transformation or degradation (e.g. combustion, digestion, composting, landfilling) and CO₂ uptake from the atmosphere through photosynthesis during biomass growth - i.e. corresponding to the carbon content of products, biofuels or above ground plant residues such as litter and dead wood. – Global Warming Potential land use and land use change (GWP-luluc): The global warming potential related to carbon uptakes and emissions (CO₂, CO and CH₄) originating from carbon stock changes caused by land use change and land use. This sub-category includes biogenic carbon exchanges from deforestation, road construction or other soil activities (including soil carbon emissions).
	Ozone Depletion	Destruction of the stratospheric ozone layer which shields the earth from ultraviolet radiation harmful to life. This destruction of ozone is caused by the breakdown of certain chlorine and/or bromine containing compounds (chlorofluorocarbons or halons), Which break down when they reach the stratosphere and then catalytically destroy ozone molecules.
	Acidification potential	Acid depositions have negative impacts on natural ecosystems and the man-made environment incl. buildings. The main sources for emissions of acidifying substances are agriculture and fossil fuel combustion used for electricity production, heating and transport.
	Eutrophication potential	The potential to cause over-fertilization of water and soil, which can result in increased growth of biomass and following adverse effects. It is split up in 3: – Eutrophication potential - freshwater: The potential to cause over-fertilization of freshwater, which can result in increased growth of biomass and following adverse effects. – Eutrophication potential - marine: The potential to cause over-fertilization of marine water, which can result in increased growth of biomass and following adverse effects. – Eutrophication potential - terrestrial: The potential to cause over-fertilization of soil, which can result in increased growth of biomass and following adverse effects.
	Photochemical ozone creation	Chemical reactions brought about by the light energy of the sun creating photochemical smog. The reaction of nitrogen oxides with hydrocarbons in the presence of sunlight to form ozone is an example of a photochemical reaction.
	Abiotic depletion potential for non-fossil resources	Consumption of non-renewable resources, thereby lowering their availability for future generations. Expressed in comparison to Antimony (Sb). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.
	Abiotic depletion potential for fossil resources	Measure for the depletion of fossil fuels such as oil, natural gas, and coal. The stock of the fossil fuels is formed by the total amount of fossil fuels, expressed in Megajoules (MJ). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.



	Ecotoxicity for aquatic fresh water	The impacts of chemical substances on ecosystems (freshwater). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Human toxicity (carcinogenic effects)	The impacts of chemical substances on human health via three parts of the environment: air, soil and water. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Human toxicity (non-carcinogenic effects)	The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Particulate matter	Accounts for the adverse health effects on human health caused by emissions of Particulate Matter (PM) and its precursors (NO_x, SO_x, NH₃)
	Resource depletion (water)	Accounts for water use related to local scarcity of water as freshwater is a scarce resource in some regions, while in others it is not. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Ionizing radiation - human health effects	This impact category deals mainly with the eventual impact on human health of low dose ionizing radiation of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator.
	Land use related impacts	The indicator is the "soil quality index" which is the result of an aggregation of following four aspects: – Biotic production – Erosion resistance – Mechanical filtration – Groundwater The aggregation is done based on a JRC model. The four aspects are quantified through the LANCA model for land use. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.



10 DETAILS OF THE UNDERLYING SCENARIOS USED TO CALCULATE THE IMPACTS

10.1 A1 – raw material supply

This module takes into account the extraction and processing of all raw materials and energy which occur upstream to the studied manufacturing process.

10.2 A2 – transport to the manufacturer

The raw materials are transported to the manufacturing site by truck (freight, lorry, >32 ton, EURO 6 or 3,5-7,5 ton, EURO 6¹).

10.3 A3 – manufacturing

This module takes into account the production process. At first, the parts for the calcium silicate damper blade and the metal tunnel body are produced as sub-assembly parts. These two parts come together in the final assembly line. Here, the following steps are performed: assembly of the damper blade, assembly of the transmission, mounting of the actuator on the damper, adding labels, end control and test. Finally, the finished and tested fire damper is packed. Production losses are also taken into account. The waste treatment of the packaging of the raw materials is modelled according to the default end-of-life scenarios of the B-EPD-PCR.

10.4 A4 – transport to the building site

Fuel type and consumption of vehicle or vehicle type used for transport	Truck >32 ton (EURO 6)	Truck 16-32 ton (EURO 6)	Truck 7,5-16 ton (EURO 6)
Distance	100 km 60% from	100 km 20% from factory to	100 km 20% from factory to

	factory to merchant	construction site 35 km 30% from merchant to installation	construction site 35 km 30% from merchant to installation
Capacity utilisation (including empty returns)	50%	50%	50%
Bulk density of transported products	Ecoinvent	Ecoinvent	Ecoinvent
Volume capacity utilisation factor	Ecoinvent	Ecoinvent	Ecoinvent

The transport of the final product from production site to building site is based on the default transport scenario for 'Finishing products – plasters', provided by the B-EPD-PCR, since this scenario best represents the real situation for fire dampers, produced by Rf-Technologies. However, in this EPD, EURO 6 trucks are used, based on the national transportation statistics for 2021 (Statbel, 2021).

The following transport steps apply:

- 20% directly to the construction site over 100 km with a 16-32 ton lorry EURO 6 (Ecoinvent v3.9.1 data record: 'Transport, freight, lorry 16-32 metric ton, EURO6 {RER}/transport, freight, lorry 16-32 metric ton, EURO6 / Cut-off, U)
- 20% directly to the construction site over 100 km with a 7,5-16 ton lorry EURO 6 (Ecoinvent v3.9.1 Cut-off data record: 'Transport, freight, lorry 7.5-16 metric ton,

¹ A truck EURO 6 is used because that type of truck is most used according to the Belgian statistics for transportation vehicles in 2021 (Statbel, 2021).



EURO6 {RER} market for transport, freight, lorry 7.5-16 metric ton, EURO6 / Cut-off, U)

- 60% to a merchant over 100 km with a >32 ton lorry EURO 6 (Ecoinvent v3.9.1 data record: '*Transport, freight, lorry >32 metric ton, EURO6 {RER} / transport, freight, lorry >32 metric ton, EURO6 / Cut-off, U*'))
- 30% from merchant to construction site over 35 km with a 16-32 ton lorry EURO 6 (Ecoinvent v3.9.1 data record: '*Transport, freight, lorry 16-32 metric ton, EURO6 {RER} / transport, freight, lorry 16-32 metric ton, EURO6 / Cut-off, U*'))
- 30% from merchant to construction site over 35 km with a 7.5-16 ton lorry EURO 6 (Ecoinvent v3.9.1 data record: '*Transport, freight, lorry 7.5-16 metric ton, EURO6 {RER} / transport, freight, lorry 7.5-16 metric ton, EURO6 / Cut-off, U*'))



10.5 A5 – installation in the building

In module A5, only the waste treatment of the packaging of the final product is considered in order to keep mass balances correct. The modelling of this EOL of the packaging is done according to the default end-of-life scenarios of the B-EPD-PCR.

Parts of the installation	Quantity	Description
Processes necessary for the installation	Not considered	
Fixation materials	Not considered	
Material losses	Not considered	
Packaging waste	1,488 kg 0,118 kg 0,059 kg	Wood Cardboard Plastics

Ancillary materials for installation			
Ancillary materials	Not considered		
Water use	Not considered		
Other resource use	Not considered		
Quantitative description of energy type (regional mix) and consumption during the installation process of the WEC® Ceiling parts	Not considered		
Waste materials on the building site before waste processing, generated by the product's installation (specified by type)	Packaging waste – cardboard: 0,118 kg	Packaging waste – plastics: 0,059 kg	Packaging waste – wood: 1,488 kg
Output materials (specified by type) as result of waste processing at the building site, e.g. of collection for recycling, for energy recovery, disposal (specified by route)	Cardboard packaging: Recycling: 95% Incineration: 5%	Plastic packaging: Recycling: 35% Incineration: 60% Landfill: 5%	Wooden pallet: Recycling: 40% Incineration: 40% Reuse: 20%
Direct emissions to ambient air, soil and water	Not considered		
Distance	Not considered		



10.6 B – use stage (excluding potential savings)

Not declared in this B-EPD

10.7 C – end-of-life

Regarding EOL, it is assumed that the waste arising from the CU-LT 200x100 fire damper with motorised actuator will first be sorted out in a sorting plant and then, according to its composition, be (partly) landfilled and/or (partly) recycled.

C1: It is assumed that no impacts are related to the demolition of the product.² The CU-LT 200x100 fire damper with motorised actuator can be easily dismantled manually, while fixings can stay in place.

C2: Transport of final product components to sorting and landfill is according to the B-EPD-PCR default scenarios.

C3-C4: Following scenarios have been used for the EOL of the different components of the final product: the metal parts are 95% recycled and 5% landfilled, while the damper blade and the plastic parts are 100 % landfilled.

Module C2 – Transport to waste processing

Type of vehicle (truck/boat/etc.)	Fuel consumption (litres/km)	Distance (km)	Capacity utilisation (%)	Density of products (kg/m ³)	Assumptions
Truck 16-32 ton EURO 5	0,260 l/km	30	50%	Eco-invent	Eco-invent
Truck 16-32 ton EURO 5	0,260 l/km	50	50%	Eco-invent	Eco-invent
Truck 16-32 ton	0,260 l/km	100	50%	Eco-invent	Eco-invent

EURO 5					
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End-of-life modules – C3 and C4

Parameter	Value (kg)
Wastes collected separately	0,000
Wastes collected as mixed construction waste	4,856
Waste for reuse	0,000
Waste for recycling	3,469
Waste for energy recovery	0,000
Waste for final disposal	1,387

² Life cycle inventories of waste treatment services: Part V "Building material disposal", page 36, table 3.20



10.8 D – benefits and loads beyond the system boundaries

In module D, following waste streams originating from the final product are considered after their end-of-waste point: the different metals (steel, copper and brass, 95% recycled). For packaging waste: wooden packaging (50% recycled), cardboard packaging (95% recycled), plastic packaging (35% recycled). The EOL waste treatments are according to the B-EPD-PCR default scenarios for the different waste materials.

QUANTITATIVE DESCRIPTION OF THE LOADS BEYOND THE SYSTEM BOUNDARIES	Production of secondary low-alloyed steel: 3,366 kg Treatment of copper scrap by electrolytic refining: 0,102 kg Production of recycled polyethylene granulates: 0,019 kg Production of recycled PET granulates: 0,001 kg The sorting and shredding of wood chips for recycling: 0,840 kg The sorting of waste paperboard for recycling: 0,130 kg
QUANTITATIVE DESCRIPTION OF THE BENEFITS BEYOND THE SYSTEM BOUNDARIES	Avoided production of primary steel: 3,366 kg Avoided production of primary copper: 0,102 kg Avoided production of primary polyethylene granulates: 0,019 kg Avoided production of primary PET granulates: 0,001 kg Avoided production of primary wood chips for wooden packaging production: 0,840 kg Avoided production of primary pulp for paper production: 0,130 kg

11 RELEASE OF DANGEROUS SUBSTANCES TO INDOOR AIR, SOIL AND WATER DURING THE USE STAGE

11.1 Indoor air

There are no emissions to indoor air caused by the CU-LT 200x100 fire damper with motorised actuator.

11.2 Soil and water

Not applicable, as this product is not in contact with (drinking) water, runoff water, seepage water and surface water.



12 DEMONSTRATION OF VERIFICATION

EN 15804+A2 serves as the core PCR.

Independent verification of the environmental declaration and data according to standard EN ISO 14025:2010

Internal ☐ External ☒

Third party verifier:

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13 TECHNICAL INFORMATION FOR SCENARIO DEVELOPMENT / APPLICATION UNIT

The information in this EPD is given for one specific fire damper with motorised actuator, produced by Rf-Technologies in its factory at Oosterzele, Belgium.

The ratio between application unit and reference flow in the EPD equals 1 (same unit in application unit as in EPD).

The fire dampers are designed and installed where air ducts penetrate fire-resistant compartment walls in order to restore the fire resistance grade of the penetrated wall and to prevent smoke propagation.

Installation can be done in different varieties, in respect with the Declaration of Performance (DoP) of the product:

- Installation in a rigid wall and floor;
- Installation in flexible wall (metal stud gypsum plasterboard wall);
- Installation in flexible wall (metal stud gypsum plasterboard wall), sealing with gypsum;
- Installation in flexible wall (metal stud gypsum plasterboard wall), sealing with mortar;
- Installation in shaft wall, sealing with gypsum;
- Installation in shaft wall, sealing with rigid rock wool boards with coating;
- Installation in shaft wall with IFW installation kit and plaster joint filler;
- Installation in flexible and rigid wall, sealing with rigid rock wool boards with coating;
- Installation in rigid floor, sealing with rigid rock wool boards with coating;
- Installation remote from the wall, sealing and insulation with rigid rock wool boards with coating;
- Installation remote from the wall + GEOFLAM plates (threaded rods and U-shaped steel profiles).

For each scenario, different materials and processes may be necessary in order to install the product. Details on the installation procedures can be found on the website of the producer through the following link: [Optimised rectangular fire damper up to 120' \(rft.eu\)](https://www.rft.eu/Optimised-rectangular-fire-damper-up-to-120/)



14 ADDITIONAL INFORMATION ON REVERSIBILITY

The reversibility of the here considered CU-LT 200x100 fire damper with motorised actuator strongly depends on the type of installation (see chapter 13). Considering the rather large variety in application, in the following table, a general qualitative description of reversibility is given for all different applications together.

Description	Type of fixing	Level of reversibility	Simplicity of disassembly	Speed of disassembly	Ease of handling (size and weight)	Robustness of material (material resistance to disassembly)	Comment
The fire damper is installed where air ducts penetrate fire-resistant compartment walls	Different types of fixation with different sealing materials	Depending on the application: – Reversible connections with light repairable damage – Reversible connections with non-repairable damage – Non reversible connections	Depending on the application: – Simple – requires the use of specific though common tools – Simple, but collecting the material is a bit more intensive	– Speed of disassembly varies from quick to slow depending on element dimensions and type and number of fixations per distance unit	– Material easy to manipulate by hand, one to two workers required depending on dimensions	– Material with a long lifespan, disassembly is possible but the material should be handled with care in order to prevent damaging it – Disassembly is possible but can cause damage to the material due to the type of assembly or fixing used	

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General information



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Project report: Life cycle assessment of a fire damper, May
2024



Verifier

Bob Roijen, SGS Intron

Date of verification: 16.09.2024

External independent verification of the declaration and data
according to EN ISO 14025 and relevant PCR documents

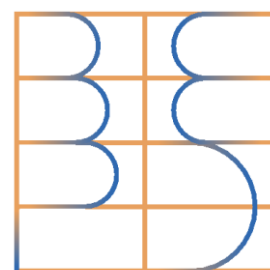
Comparing EPDs is not possible unless they are conform to the same PCR and taking into account the building context.
The program operator cannot be held responsible for the information supplied by the owner of the EPD nor LCA practitioner.



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